

Physics-Informed Differentiable Learning for High-Order Nonlinear Reduced-Order Models of Hyperbolic PDEs

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1 Introduction

Reduced-order modeling (ROM) of parameterized, time-dependent PDEs is critical for making large-scale simulations tractable. Classical projection-based methods that represent the solution as a linear combination of prescribed modes are provably inefficient for hyperbolic problems: the Kolmogorov n -width of advection-dominated solution manifolds decays slowly, forcing impractically large linear subspaces [2]. Nonlinear manifold approaches address this by learning compact latent representations via deep autoencoders [4], but they introduce a new set of challenges.

Standard data-driven nonlinear ROMs decouple the surrogate from the governing physics after data generation. Autoregressive propagators—RNNs, LSTMs, or neural ODEs—are trained purely on snapshot sequences and carry no knowledge of the underlying differential operator. This disconnect causes the latent trajectory to drift from the true physics, producing severe error accumulation over long rollout horizons and poor generalization to unseen parameter regimes. A further, less-discussed issue is geometric: the nonlinear reduction map must satisfy strict regularity conditions for the reduced equations to remain well-posed.

Formally, the set of all reduced solutions defines an n -dimensional manifold $\mathcal{M}$ immersed in the Hilbert space H of the PDE [1]. The reduction map $\hat{u} : q \mapsto \hat{u}(\cdot, q)$ must satisfy:

- **Injectivity and differentiability:** $\hat{u}$ is injective and C^1 with respect to the latent coordinates q .
- **Spatial regularity:** for every $q \in \Omega$, the field $\hat{u}(\cdot, q)$ has sufficient spatial smoothness to be evaluated in the PDE.
- **Immersion (full-rank tangent space):** the partial derivatives $\partial_{q_1}\hat{u}, \dots, \partial_{q_n}\hat{u}$ are linearly independent, spanning a full-rank tangent space $T_{\hat{u}}\mathcal{M}$.

When the immersion condition is violated—a phenomenon known as *tangent space collapse*—the Riemannian metric tensor $G_{ij} = \langle \partial_{q_i}\hat{u}, \partial_{q_j}\hat{u} \rangle_H$ becomes singular [6], rendering the reduced dynamics ill-posed and destroying training stability.

We propose an end-to-end differentiable framework that simultaneously addresses both the physics-fidelity and geometric-stability problems. Rather than discarding the high-fidelity solver after dataset generation, we embed the high-order DG spatial operator directly into the training loss. Using the Tesseract framework [3] for cross-language differentiable orchestration, we couple a PyTorch coordinate-based neural network to a Julia/Trixi.jl DG solver [5] such that exact analytical

Vector-Jacobian Products (VJPs) of the DG flux propagate loss gradients upstream without truncation error. A Gram-matrix regularizer simultaneously enforces full-rank tangent spaces throughout training, preventing collapse and enabling stable long-horizon rollout.

2 Methodology

2.1 Architectural Formulation

The framework learns a continuous nonlinear solution manifold paired with a parameterized latent propagator through two feed-forward networks:

- **Manifold network** D_θ : maps physical coordinates $x \in \Omega$ and the latent state $q(t) \in \mathbb{R}^n$ to the reconstructed field $\hat{u}(x, t) = D_\theta(x, q(t))$. Crucially, D_θ is an implicit neural representation (INR) evaluated at GLL quadrature nodes, ensuring consistency with the DG discretization.
- **Temporal dynamics network** Ψ_ψ : predicts the continuous-time latent velocity $\dot{q}(t)$ conditioned on the current state q and the PDE parameter β .

The learnable parameters are cleanly partitioned into spatial manifold parameters θ and temporal parameters ψ , each governed by dedicated loss terms.

2.2 Physics-Informed Loss Formulation

Training minimizes a composite objective that balances reconstruction accuracy, physical fidelity, geometric stability, and temporal tracking.

Manifold reconstruction loss $\mathcal{L}_{\text{man}}$. Measures the quadrature-weighted L^2 discrepancy between FOM snapshots and the manifold reconstruction, ensuring the manifold fits the training data:

$$\mathcal{L}_{\text{man}}(\theta) = \frac{1}{B} \sum_{b=1}^B \|\hat{U}^{(b)} - U_{\text{FOM}}^{(b)}\|_W^2. \quad (1)$$

Mechanistic tangent loss $\mathcal{L}_{\text{tan}}$. By the chain rule, the temporal rate of change on the manifold satisfies $\hat{u}_t = J_D(q) \dot{q}$, where $J_D = \partial D_\theta / \partial q$ is the manifold Jacobian. Projecting the exact DG flux $F(\hat{u})$ —evaluated natively in Julia—onto $T_{\hat{u}}\mathcal{M}$ via regularized least-squares yields the optimal target dynamics:

$$\dot{q}^* = (G(q) + \delta_{\text{reg}} I_n)^{-1} J_D(q)^\top W F(\hat{u}), \quad G(q) = J_D(q)^\top W J_D(q). \quad (2)$$

The tangent loss then penalizes the instantaneous projection residual:

$$\mathcal{L}_{\text{tan}}(\theta) = \frac{1}{B} \sum_{b=1}^B \frac{\|J_D(q^{(b)}) \dot{q}^{*(b)} - F(\hat{U}^{(b)})\|_W}{\|F(\hat{U}^{(b)})\|_W + \epsilon}. \quad (3)$$

This loss forces θ to shape the manifold so that the DG physics can be faithfully represented within the learned tangent space, directly coupling the neural representation to the numerical scheme.

Gram matrix regularization $\mathcal{L}_{\text{reg}}$. To prevent tangent space collapse, we penalize the deviation of the metric tensor from the identity:

$$\mathcal{L}_{\text{reg}}(\theta) = \frac{1}{B} \sum_{b=1}^B \|G(q^{(b)}) - I_n\|_F^2. \quad (4)$$

Minimizing this term keeps the tangent basis vectors mutually orthonormal, ensuring G remains well-conditioned and the reduced dynamics well-posed.

Temporal dynamics loss $\mathcal{L}_{\text{temp}}$. Trains Ψ_ψ to track the optimal tangent velocity $\dot{q}^*$:

$$\mathcal{L}_{\text{temp}}(\psi, \theta) = \frac{1}{B} \sum_{b=1}^B \frac{\|\Psi_\psi(q^{(b)}, \beta^{(b)}) - \dot{q}^{*(b)}\|_2}{\|\dot{q}^{*(b)}\|_2 + \epsilon}. \quad (5)$$

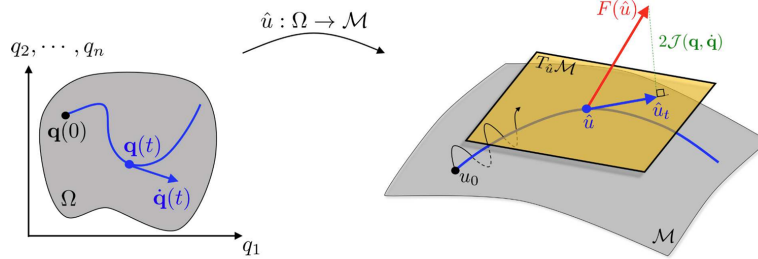


Figure 1: Geometric illustration of the RONS framework [1]. The reduced solution evolves on $\mathcal{M}$, with $q(t)$ tracing a path in Ω (left). The DG flux $F(\hat{u})$ (red vector) is projected onto the tangent space $T_{\hat{u}}\mathcal{M}$ (yellow plane), yielding the optimal latent velocity $\dot{q}^*$.

2.3 Cross-Language Differentiability and Two-Phase Training

The core technical contribution is the exact gradient pathway between the Julia DG solver and PyTorch’s autograd engine. Via the Tesseract API, the DG operator is exposed as a differentiable function: the forward call evaluates $F(\hat{U})$ natively in Julia, while the backward call computes the analytical discrete adjoint $(J_F)^\top v = (\partial F / \partial u)^\top v$ via the block-tridiagonal sparsity structure of the LLF flux Jacobian. This guarantees that gradients flowing through the DG operator carry no finite-difference truncation error, a property that is non-negotiable for the tangent loss to provide a meaningful training signal. The cross-language pipeline is illustrated in Fig. 2.

To prevent trivial manifold collapse—where θ warps the latent geometry purely to minimize $\mathcal{L}_{\text{temp}}$ at the expense of physical fidelity—we employ a two-phase staged training strategy. **Phase 1 (decoupled):** the latent state q is detached before being passed to Ψ_ψ , so temporal gradients update only ψ while θ is driven exclusively by the geometric and tangent losses. **Phase 2 (joint fine-tuning):** the full computation graph is reconnected, allowing temporal gradients to gently refine the spatial manifold under the combined objective. The full procedure is given in Algorithm 1.

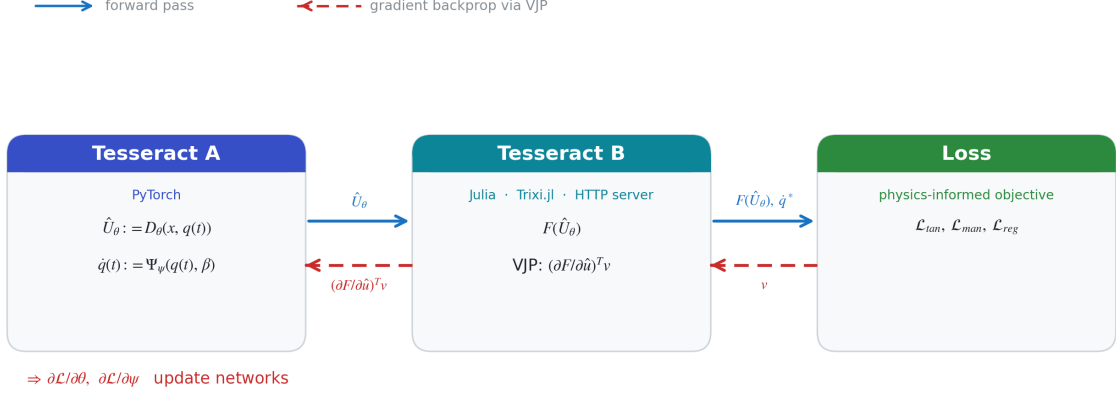


Figure 2: Cross-language differentiable pipeline. *Forward pass* (blue): PyTorch passes $\hat{U}_\theta$ to the Julia DG server, which returns $F(\hat{U}_\theta)$. *Backward pass* (red dashed): the analytical VJP $(\partial F / \partial \hat{u})^\top v$ flows back into PyTorch autograd, updating θ and ψ via exact gradients.

3 Numerical Experiments

3.1 Setup

We evaluate on the 1D inviscid Burgers equation $\partial_t u + \frac{1}{2} \partial_x (u^2) = 0$ with the periodic N-Wave initial condition:

$$u(x, 0) = \mu_1 \sin(2\pi x) + 0.3 \sin(4\pi x + \mu_2). \quad (6)$$

This configuration generates multi-harmonic wave interference and steepening shear layers—a stringent test for nonlinear manifold methods. The parameter μ_1 governs amplitude and is sampled at 50 uniformly spaced points over $\mu_1 \in [0.5, 1.5]$, with μ_2 fixed. FOM snapshots are generated by Trixi.jl using $K = 100$ DG elements with polynomial degree $p = 3$, periodic boundary conditions, and integration to final time $T = 0.5$.

3.2 Results

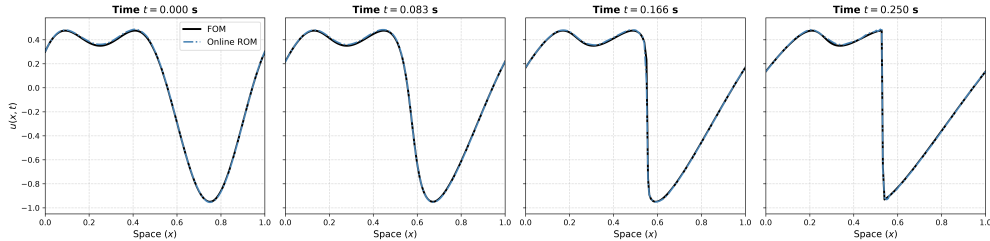


Figure 3: Spatiotemporal evolution of the N-Wave for $\mu_1 = 0.57$.

Fig. 3 shows the online ROM rollout to $T = 0.25$ for $\mu_1 = 0.57$. The learned manifold correctly resolves the wave steepening and multi-harmonic features, achieving a relative L^2 error of

Algorithm 1 Physics-Informed Two-Phase Differentiable Training

Input: Dataset $\mathcal{U}_{\text{tr}}$, Tesseract session $\mathcal{S}_{\text{tess}}$, transition epoch e_{joint} , weights $\lambda_m, \lambda_t, \lambda_r, \lambda_{\text{temp}}$

- 1: Initialize manifold parameters θ and temporal parameters ψ
- 2: **for** epoch = 1 **to** N_{epochs} **do**
- 3: **for all** mini-batches $\{U, \beta\} \subset \mathcal{U}_{\text{tr}}$ **do**
- 4: $\hat{U}, J_D \leftarrow D_{\theta}(x_{\text{grid}}, q), \partial D_{\theta}/\partial q$
- 5: $F(\hat{U}) \leftarrow \mathcal{S}_{\text{tess}}.\text{flux}(\hat{U}, \beta)$ // Forward DG flux via Tesseract
- 6: $\dot{q}^* \leftarrow (J_D^{\top} W J_D + \delta I)^{-1} J_D^{\top} W F(\hat{U})$ // Tangent projection
- 7: $\mathcal{L}_{\text{spatial}} \leftarrow \lambda_m \mathcal{L}_{\text{man}} + \lambda_t \mathcal{L}_{\text{tan}} + \lambda_r \mathcal{L}_{\text{reg}}$
- 8: **if** epoch $\leq e_{\text{joint}}$ **then** // Phase 1 — decoupled
- 9: $\dot{q}_{\text{pred}} \leftarrow \Psi_{\psi}(\text{detach}(q), \beta)$
- 10: Update θ via $\nabla_{\theta} \mathcal{L}_{\text{spatial}}$ (exact VJPs through Tesseract)
- 11: Update ψ via $\nabla_{\psi} \mathcal{L}_{\text{temp}}$
- 12: **else** // Phase 2 — joint fine-tuning
- 13: $\dot{q}_{\text{pred}} \leftarrow \Psi_{\psi}(q, \beta)$
- 14: $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{spatial}} + \lambda_{\text{temp}} \mathcal{L}_{\text{temp}}$
- 15: Update θ, ψ jointly via $\nabla_{\theta, \psi} \mathcal{L}_{\text{total}}$
- 16: **end if**
- 17: **end for**
- 18: **end for**

approximately 4×10^{-2} against the FOM. Accuracy degrades for longer simulation horizons, an expected consequence of accumulated tangent projection error when the solution approaches shock formation. Improving long-horizon stability—through adaptive latent dimensionality or time-local manifold recentering—is an immediate direction for future work.

References

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